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After a Decade Increased by 650,000 in 2022

Robert Warren View all authors and affiliations

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Contents

- Executive Summary
- Background
- Continent or Area of Origin
- Country of Origin
- State of residence
- Undocumented Population Growth Versus Apprehensions by DHS
- Summary
- Declaration of Conflicting Interests
- Funding
- Footnotes
- Appendix
- References

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- Share options
- Information, rights and permissions
- Metrics and citations
- Figures and tables

Executive Summary

This report describes estimates of the undocumented population residing in the United States in 2022 compiled by the Center for Migration Studies of New York (CMS). The estimates are based on data collected in the American Community Survey (ACS) conducted by the US Census Bureau (Ruggles et al. 2023). The report finds that the undocumented population grew from 10.3 million in 2021 to 10.9 million in 2022, an increase of 650,000. The increase reverses more than a decade of gradual decline. The undocumented populations from 10 countries increased by a total of 525,000: Mexico, the Dominican Republic, and India; El Salvador, Guatemala, and Honduras in Central America; and Brazil, Colombia, Ecuador, and Venezuela in South America. The undocumented population in Florida increased by about 125,000 in 2022, Texas increased by 60,000, New York by 50,000, and Maryland by 45,000. The report explains why undocumented population growth is much less than the number of apprehensions by DHS. Finally, the Appendix provides a detailed description of the CMS methodology.

- After remaining at or near zero growth from 2010 to 2021 (Warren 2023), the US undocumented population increased by 650,000 in 2022.
- The largest population gains in 2022 were for Central America (205,000), South America (200,000), and Asia (140,000).
- From 2015 to 2022, the undocumented population from Mexico declined by 1.3 million; in the same period, the combined population from Central and South America increased by 1.2 million.
- The undocumented population from Asia declined by 115,000 from 2015 to 2021 and then increased by 140,000 in 2022.
- California, Texas, Florida, New York, and New Jersey had the largest undocumented populations in 2022. The total population in those five states increased by 300,000 in 2022.
- The combined undocumented population from El Salvador, Guatemala, and Honduras was undercounted by almost 400,000 in 2022. Correcting this data accounted for more than half the estimated increase in the undocumented population in 2022.

Background

The rapid increase in annual apprehensions and expulsions of migrants along the U.S.-Mexico border that began in 2019 has dominated media attention on migration for the past four years. That, in turn, has focused attention on the prospect of large annual increases in the undocumented population. Unfortunately, ACS data needed to monitor changes in the population are not available until about a year after the ACS survey is completed. Most organizations require six months or more to compile estimates, which means in most cases estimates of change are about two years out of date. The unique CMS methodology made it possible to derive these estimates less than two months after the release of 2022 ACS data.

This report describes changes in the US undocumented population by country of origin and state of residence since 2018 with special emphasis on changes from 2021 to 2022. The estimates were derived from ACS data by selecting noncitizens that arrived after 1982,[1] removing those determined to be legal residents based on characteristics reported in the survey, and further reducing the number using the techniques detailed in Appendix B.

After estimating the number of u S. For most countries of origin, the undercount rate varies ustments for El Salvador, Guatemala, and Honduras. The t

An important issue to be resolve d Honduras in the ACS in 2020 to 2022 were the result of l Appendix A confirms that the relatively low counts for these c e appropriate.

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Continent or Area of

The undocumented population d ion from Mexico declined while the population from Centr l to grow, but the population from Mexico stopped declining, om an annual average of 30,000 from 2018 to 2021 to an

Table 1. Estimated Undoc

	Estimated undocumented population (July)					Average change	Change
Area of origin	2018	2019	2020	2021	2022	2018–2021	2021–2022
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
US total	10,565	10,350	10,210	10,290	10,940	−90	650
Mexico	5,110	4,750	4,445	4,425	4,485	−230	60
Central America	1,740	1,925	2,120	2,230	2,435	165	205
Caribbean	445	435	435	425	465	−5	40
South America	745	795	820	835	1,040	30	200
Europe	310	290	285	305	310	–	5
Asia	1,765	1,715	1,650	1,650	1,790	−40	140
Africa	375	370	380	355	345	−5	−15
All other	75	65	80	60	70	−5	10

Note: Rounded to 5,000s.
Source: Center for Migration Studies. – zero or rounds to zero.

Figure 1 shows trends in the total undocumented population from 2015 to 2022 and trends for Mexico compared to Central and South America combined. Over that period, the population from Mexico declined by 1.3 million while the population from Central and South America combined increased by 1.2 million.



Figure 1. Estimated Undocumented Population, 2015 to 2022: Total, Mexico, and Central and South America.

Country of Origin

Table 2 shows annual estimates of the population from 2018 to 2022 for the top 15 countries of origin. The populations from El Salvador, Guatemala and Honduras increased steadily from 2018 to 2022. The pattern changed for most of the other countries in Table 2, with considerably larger increases in 2022 than in the earlier years. The population from Venezuela increased more in 2022 than it had in the previous three years combined (Table 2). The populations from Brazil, Dominican Republic, and Colombia also increased in 2022.

Table 2. Estimated Undocumented Population From the Top 15 Countries: 2018 to 2022.

State of residence	Estimated undocumented population (July)					Average change,	Change,
	2018 (1)	2019	2020	2021	2022	2018-'21	2021-'22
US total	10,565						
Mexico	5,110						
Guatemala	575						
El Salvador	690						
India	620						
Honduras	385						
China	360						
Venezuela	170						
Brazil	140						
Dom. Rep.	185						
Colombia	145						
Philippines	195						
Ecuador	110	105	95	105	140	–	35
Korea	160	140	130	120	130	−15	10
Haiti	115	115	115	110	115	−5	5
Nicaragua	55	50	45	50	95	–	45
All other	1,550	1,525	1,510	1,485	1,530	–	10

Source: Center for Migration Studies. – zero or rounds to zero.
Note: Rounded to 5,000s. Ranked by population in 2022.

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After 2020, Ecuador changed from a country that had 10 years of steady decline in the undocumented population to a country with rapid growth. The undocumented population from Ecuador was 175,000 in 2010, and it declined to 95,000 in 2020, a drop of 80,000 in ten years. As the data in Table 2 show, the population changed course after 2020, increasing by 45,000 from 2020 to 2022.
According to one analysis, the third significant wave of Ecuadorian emigration since the 1980s began in 2019, driven by the devastating economic impact of the COVID-19 pandemic, increased unemployment, poverty, rising crime, extortion, corruption, the growing influence of narcotraffickers, and “the government’s inability to address these issues” (Jokisch 2023). The current era of emigration has been characterized by increased irregular migration, and greater migration by unaccompanied children and family units, based on the belief that migrants with children can enter the United States (ibid.).
The undocumented population from India increased from 460,000 in 2015 to more than 600,000 in 2018 (Figure 2). It held steady from 2018 to 2020 and began to increase again after 2020. The undocumented population from other major Asian sending countries declined or remained stable from 2015 to 2022.



Figure 2. Estimated Undocumented Population From Selected Asian Countries: 2015 to 2023.

State of residence

Some states gained more than others, but the increase in the undocumented population in 2022 affected every state. Table 3 shows the increase in the undocumented population in the top 15 states in 2022. All the numbers in Table 3 have been rounded to 5,000s. The population in each of the 15 states increased by more in 2022 than it did, on average, from 2018 to 2021. The "all other" category – 35 States and DC – lost about 25,000, on average, from 2018 to 2021; those countries increased by 165,000 in 2022 (Table 3).

Table 3. Estimated Undocumented Population Residing in the Top 15 States: 2018 to 2022.

All numbers rounded to 5,000s

State of residence	Estim… 2018 (1)						
US total	10,565						
California	2,310						
Texas	1,795						
Florida	755						
New York	685						
New Jersey	415						
Illinois	450						
Georgia	345						
North Carolina	300						
Washington	270						
Maryland	215						
Arizona	260	260	225	235	250	−10	15
Virginia	265	260	245	250	245	−5	–
Massachusetts	180	175	180	180	215	–	35
Pennsylvania	185	155	180	175	180	−5	5
Colorado	160	145	150	155	165	–	5
All other	1,970	1,935	1,850	1,895	2,055	−25	165

Note: Ranked by population in 2022.
Source: Center for Migration Studies. – zero or rounds to zero.

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The undocumented population in Florida increase by 125,000 in 2022 after growing by 20,000 annually from 2018 to 2021 (Table 3). The countries of origin with the largest population growth in Florida in 2022 were Venezuela (45,000), Guatemala (25,000), Nicaragua (20,000), El Salvador (15,000), Columbia (15,000), and Haiti (15,000). The undocumented population from Mexico living in Florida declined by 15,000.
Table 4 shows the top 15 states in 2022 and the country that had the largest increase in undocumented residents in each state. Mexico was the largest contributor in five states — Iowa, Oklahoma, Oregon, Washington, and Arizona (Table 4). Undocumented migrants from Guatemala increased the most in New York, California, and Georgia. Michigan, North Carolina, and Illinois had the largest increases from India.

Table 4. States With the Largest Increase in Undocumented Residents in 2022.

State	Pop. in 2021 (1)	Pop. in 2022 (2)	Increase (3) = (2)–(1)	Percent increase (4) = (3)/(1) × 100	Country that had the largest population increase in each state.	
US total	10,290	10,940	650	6	Country	Change
Florida	810	935	126	16	Venezuela	44
Texas	1,790	1,850	59	3	El Salvador	24
New York	620	670	52	8	Guatemala	16
Maryland	240	285	46	19	El Salvador	16
California	2,160	2,200	38	2	Guatemala	15
Massachusetts	180	215	35	19	Brazil	15
Georgia	310	340	32	10	Guatemala	10
Iowa	35	65	28	83	Mexico	8
New Jersey	470	495	27	6	Ecuador	14
Oklahoma	70	95	25	35	Mexico	20
Oregon	95	115	24	26	Mexico	10
Washington	275	290	18	6	Mexico	6
Michigan	100	115	17	18	India	7
Connecticut	105	125	17	16	Honduras	7
Indiana	100	115	16	16	Honduras	3
Arizona	235	250	16	7	Mexico	20
North Carolina	310	325	15	5	India	6
Illinois	415	430	13	3	India	7
Utah	85	100	13	14	Venezuela	7
Wisconsin	75	85	11	15	Nicaragua	5

Note: Numbers in thousands. States ranked by population increase.
Source: Center for Migration Studies.

Undocumented Population Growth Versus Apprehensions by DHS

The question might arise: How could the undocumented population grow by “only” 650,000 in 2022 when DHS apprehensions exceeded 2.5 million? The answer is straightforward: Demographically, apprehensions and population growth are very different concepts. Total apprehensions essentially are the sum of failed attempts to enter, while undocumented population growth is the net result of successful entries across the border, overstays of temporary admissions, voluntary emigration, removal by DHS, adjustment to lawful status, and death.

Figure 3 shows annual growth of the undocumented population and DHS apprehensions from 2010 to 2022. Apprehensions were stable at about 750,000 from 2010 to 2018. Beginning in 2019, apprehensions increased annually in stair-step fashion, except in 2020, the peak year of the COVID-19 pandemic (Figure 3). The orange bars show annual change in the size of the undocumented population. The data used to construct Figure 3 show that 6.5 million apprehensions occurred from 2010 to 2018; the undocumented population *declined* by 1.1 million during that period. As the data on Figure 3 show, any link between the number of apprehensions and po

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Figure 3. DHS Apprehensions and Undocumented Population Growth: 2010 to 2022.
Source: DHS apprehensions, Yearbook Table 33; population estimates, DHS.

Kerwin and Warren (2023) summarized the reasons why apprehensions by DHS do not translate directly into undocumented population growth: “[S]ome migrants are apprehended multiple times, some are trying to return to a permanent residence in the United States after a visit to their communities of origin, some are seasonal workers, and some are coming temporarily to visit family. None of these cases would add a new resident to the undocumented population . . .The fact that the Border Patrol *prevents* most attempted entries has not received wide media coverage. In 2017, DHS estimated that it interdicted 80 percent of attempted entries in the 2014 to 2016 period” (citation omitted).

The numbers arriving illegally across the border and the numbers overstaying temporary visas each year are offset by the numbers leaving the undocumented population. From 2011 to 2021, an annual average of more than 500,000 left the undocumented population through voluntary emigration, removal by DHS, adjustment to legal status, or death (Warren 2023, Table 2).

Summary

In 2022, the undocumented population in the United States increased by 650,000, the largest increase since 2001, when the population increased by slightly more than one million. The population grew rapidly because the population from Mexico stopped declining, undocumented migration from Central America continued to increase, arrivals from South America, particularly Venezuela, were higher than in previous years, and undocumented migration from India continued to grow.

The large population growth that began in 2022 can be expected to continue as long as the US Congress is unable to address the immigration system in a bipartisan manner. The last time the undocumented population had a large growth spurt was in 1999 to 2001; the population increased by 1.8 million in that two-year period. With bipartisan support, the size of the Border Patrol was doubled in the Bush administration and doubled again in the Obama administration. What followed was a dramatic decline in apprehensions and in undocumented population growth (Warren and Warren 2013). From 2010 to 2021, the total undocumented population declined by about 1.4 million (Warren 2023).

Large-scale illegal entries undermine the rule of law. Yet as a recent CMS report illustrates, so does underinvestment in the immigration court system (Kerwin and Millet 2023). As it stands, US immigration courts labor under a backlog of nearly 2.5 million pending cases, following an increase of 672,000 cases in FY 2023 (US Department of Justice, Executive Office for Immigration Review, 2023). Court backlogs delay the adjudication of meritorious asylum cases and the removal of persons with specious claims. The 4.1 million immigrant visa backlog (US Department of State 2021) also undermines the rule of law. It prevents persons from securing visas for which they are eligible — in some cases for decades. Many in the visa backlog languish in the court backlog as well. These backlogs can be easily reversed — from a technical standpoint — through legislative and administrative fixes. Yet the larger need is for an integrated approach to reform of the US immigration system as a whole. Piecemeal and narrow fixes will not suffice.

Declaration of Conflicting Interests

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Footnotes

1 Data were selected for noncitizens that arrived after 1981 because undocumented residents that arrived before that date would have acquired legal status under the provisions of the Immigration Reform and Control Act of 1986 (IRCA).
2 Undercount rates vary by country based on year of entry, with more recent arrivals receiving a larger adjustment for undercount. Countries with relatively more recent arrivals have higher
3 Appendix B provides a detaile ulation.
4 Undercount for these countries stration's widely publicized efforts to remove undocumented
5 The adjusted figure for 2020 n
6 The term logical edit refers to t ere assigned to the legal category if they worked in occup immediate relatives of US citizens, received public benefits age 60 or older at entry.
"Edited population" refers to AC
7 The countries and areas are the
8 The population and componen miniscule effect on the estimates.
9 For the estimation procedure s
https://www.dhs.gov/sites/defaul IS%201990%20to%202000.pdf

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Appendix

Appendix A

Extra Adjustment for Undercount for El Salvador, Honduras, and Guatemala

Additional adjustments for undercount were necessary for the ACS data for El Salvador, Guatemala, and Honduras. "Additional" means in addition to the usual undercount rates applied to undocumented residents of each country.[3] The total addition for undercount for these three countries was about 395,000 in 2022. The adjustment was made separately for each country, but the three countries are summed up in this Appendix to simplify the explanation of the methodology. "Likely undocumented residents" in Figure A1 refers to the noncitizen population that arrived from these three countries after 1981. Noncitizens judged to be legal residents based on characteristics reported in the survey were removed from the data.



Figure A1. Likely Undocumented Residents From EI Salvador, Honduras, and Guatemala (combined) Counted in the ACS.

The figures on the red line in Figure A1 are the numbers counted in the ACS. The figures on the black line for 2020 to 2022 were derived by extrapolating growth rates from 2017 to 2019 forward. The adjustments to the data in Figure A1 assume that the sharp decline in the population in 2020, and continuing to 2022, occurred because of undercount in the ACS[4] and not because of the unlikely alternative — that more than 500,000[5] undocumented residents emigrated to these countries in 2020.
It is logical to assume that the sharp drop in the numbers counted in the ACS from these countries in 2020 to 2022, shown on the red line in Figure A1, are the result of undercount and not the result of mass emigration to the three Central American countries. Because of the importance of the estimates for 2022, at a time in which apprehensions are at a very high level, we conducted an additional demographic analysis to confirm that the pattern observed in Figure A1 was the result of undercount in the ACS and not because of large-scale emigration to these Central American countries.
Figure A2 summarizes the results of an analysis of changes in the undocumented immigrant cohort that arrived from 1982 to 2013. This cohort was chosen for analysis because *all* the migrants were in the ACS in all five years of this analysis, and they were relatively permanent residents who were not as likely as recent arrivals to emigrate.

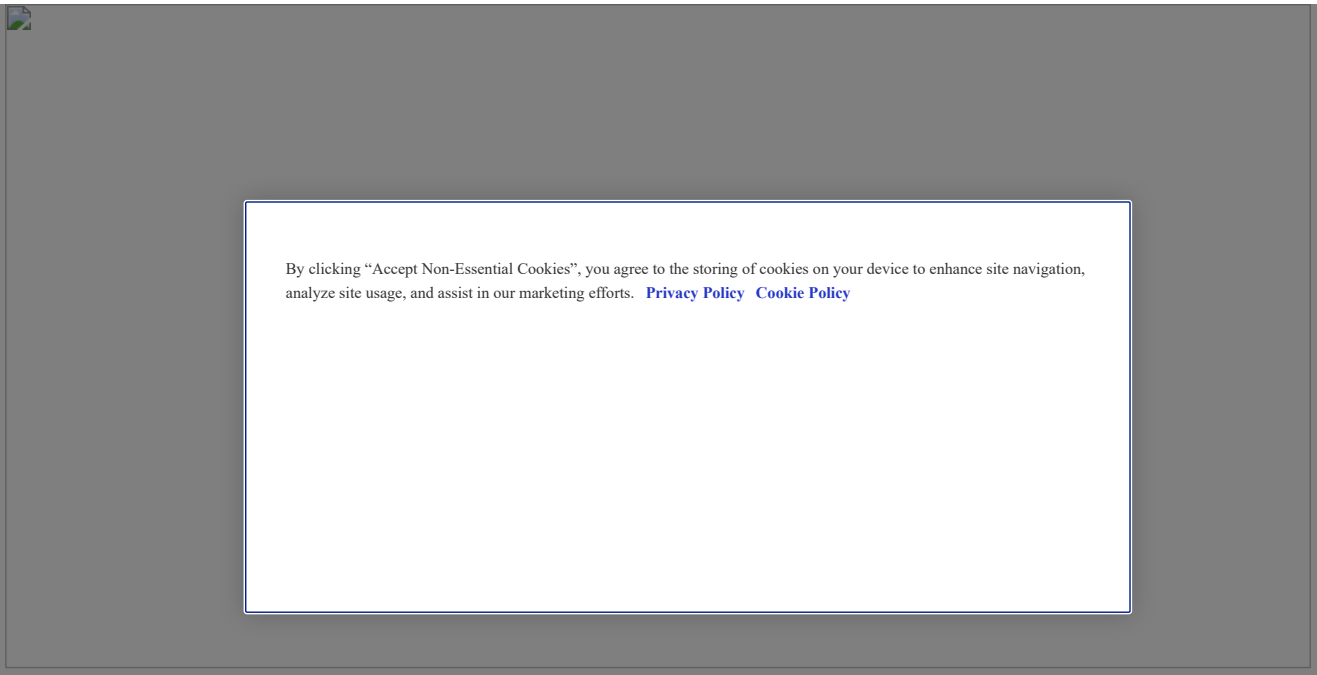


Figure A2. Change in the 1982 to 2013 Entry Cohort From EI Salvador, Guatemala and Honduras (combined).

The data points on the black line for 2018 and 2019 show that the population declined slightly (Figure A2). This is as expected because the population that arrived from 1982 to 2013 can only decrease as members of the cohort leave the undocumented population through voluntary emigration, removal by DHS, adjustment to legal status, or death. The data points on the black line for 2020 to 2022 continue the expected small decline in this cohort. The figures on the red line are the numbers enumerated in the ACS. The differences between the black and red lines are a measure of undercount in the ACS.
The similarity of the patterns in these two graphs confirm that the relatively low counts of undocumented residents from these three Central American countries in 2020, 2021, and 2022 were the result of undercount in the ACS and not the result of large-scale emigration. In addition, this analysis provides empirical support for the estimates of undercount in these three countries from 2020 to 2022.

Appendix B

CMS Methodology for Deriving Annual Estimates of the Undocumented Population, Beginning in 2010

Deriving US Estimates of Undocumented Residents Counted in the 2010 ACS

The first step was to derive estimates of the undocumented population *counted in the 2010 ACS*, by country of origin. The estimate for each country had to fit within two parameters:
First: The total undocumented population counted in 2010 ACS had to sum to 10,850,000, an estimate published in Warren and Warren (2013).
Second: For each country/area, the population counted in 2010 ACS could not exceed the “edited” population. Edited population, as used here, refers to noncitizens in the ACS that arrived after 1981, minus those judged to be legal residents based on the logical edits.[6]
Within those two parameters, CMS used the residual technique to derive *provisional* estimates for each of the 146 countries/areas.[7] Next, independent databases were used to refine the individual country estimates. This made the 146 countries/areas sum to 10,850,000 and generated the best empirical relationships between the countries.
After this step was completed, an estimate of undocumented residents counted in the in 2010 ACS was available for each of the 146 countries/areas; the sum was 10,850,000, consistent with the Warren and Warren (2013) estimates. A more detailed description of the CMS method of estimation for 2010 is provided in Warren (2014).

Assigning Legal Status to Respondents in the 2010 ACS

The estimates of the undocumented population described in step 1 were the numerators of a set of ratios (also referred to as selection ratios) used to assign legal status to respondents in the 2010 ACS. The term ratio refers to the undocumented population divided by the “edited” population. The ratios for each of the 146 country/areas were used to randomly select cases from the edited ACS microdata.

Deriving Annual Estimates of Undocumented Counted in the ACS, 2011 to 2016

The ratios compiled in Step 2 were used, along with annual ACS data, to derive estimates of the undocumented population from each country/area (listed in IPUMS) for each year from 2011 to 2016. The procedure used each year was straightforward: the edited population from each country/area was multiplied by that country/area’s ratio derived for 2010, and the result was randomly selected from the edited population.

Updating the Population Estimates, and Thus the Selection Ratios, to 2017

The rationale for using the *same* ratio to derive the estimates from 2011 to 2016 was that the population numbers for 2010, and thus the 2010 ratios, are the *cumulative result* of legal and undocumented entries over the entire 28-year period from 1982 to 2010. If the proportion of undocumented to legal entries in 2011 deviated from that long-term trend, the effect on the ratio, and thus the estimate, would likely be small. Examination of the annual ACS data for noncitizens, as well as annual DHS data for LPRs admitted, shows that arrivals and population trends for nearly all countries tend to be stable over time. Finally, the (presumably) minor changes that might occur each year in the ratios for each of the 145 countries or areas would offset each other to some extent, producing stable estimates of the total undocumented population over time.

The justification in the previous paragraph for using the same country-specific ratios from year to year is likely to be valid in the short run. However, in the longer term it is important to have an empirically-based procedure to validate that assumption and to update the estimates, and therefore the ratios — in this case, after seven years. For the 2017 estimates, the ratios for 116 countries were revised, as needed, based on change in the estimated legally resident population, derived from DHS administrative data and estimates of deaths and emigration.[8] Table A1 shows how the ratio for Mexico was revised in 2017.

Table A1. Method Used to Update the Selection Ratio for Mexico From 2010 to 2017.

Numbers in thousands, rounded independently.

1.	
2.	
3.	
4 = 3/2	
5 = 1–3	
6.	
7.	
8.	
9.	
10.	
11 = 5 + 6 – 7 + 8 – 9 – 10	
12.	
13 = 12 – 11	
14.	5,265 Edited population in 2017 compiled by CMS
15 = 13/14	.946 Revised ratio for 2017

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Source: Center for Migration Studies.

a
Annual rate of 1.0%, derived from 2010 to 2017 ACS data. Method of estimation of emigration rates available upon request.

As Table A1 shows, the estimated *legally resident noncitizen population* is the critical component in this method of revising the ratios. Data are readily available for estimating the components of change of this population – items 6 through 10 in Table A1. The likely margin of error for the total legally resident population in 2017 (Item 11) is small because the largest components of change (LPRs and naturalizations) are well-measured administrative data.

The selection ratio for Mexico changed from 0.956 to 0.945. The revised ratio *reduced* the 2010-based estimate of the undocumented population from Mexico by 53,000, or 1.0 percent. This decline occurred because the undocumented population from Mexico dropped steadily after 2010, while the legally resident population increased (Table A1). The revised ratios raised the estimated undocumented population for some other countries and reduced the population for other countries. For the total population, the net result of updating the ratios in 2017 was less than ½ percent. That is, the updated ratios changed the total population in 2017 by less than ½ percent compared to estimates for 2017 derived using the original 2010 ratios.

Overall, this exercise has shown that keeping the ratios constant for seven years introduced little error into estimates for the leading sending countries and for the total estimated undocumented population. The analysis and revisions also show that adequate information is available to make any necessary adjustments.

Adjusting for Undercount of Undocumented Residents in the ACS

The CMS estimates are adjusted for undercount using an approach like the one described by Warren and Warren (2013). The procedure is analogous to methods used by (former) INS to derive estimates for 2000.[9] In both cases, the undercount rate is relatively higher for recent arrivals than for those who have been in the US the longest. As noted above, in the Warren and Warren (2013) estimates, the number counted in the 2010 ACS was 10,850,000, and the population adjusted for undercount was 11,725,000. Within those two constraints, annual undercount rates were derived so that (1) undercount would be highest in the most recent year of entry, and (2) the undercount rate would decline to about 2% in 1982 (the earliest year of entry of undocumented residents).

Specifically, for the 2010 CMS estimates, those that entered in 2010 and 2009 were adjusted for undercount by 13.2 percent and the rate dropped by 7 percent each year, falling to 1.9 percent for those who entered in 1982. Adjustments for undercount were made by re-weighting the microdata by single year of arrival from 1982 to the survey date. The CMS undercount assumptions are generally consistent with undercount rates for the Hispanic male population counted in the ACS as measured by the Census Bureau (Jensen et al., 2015).

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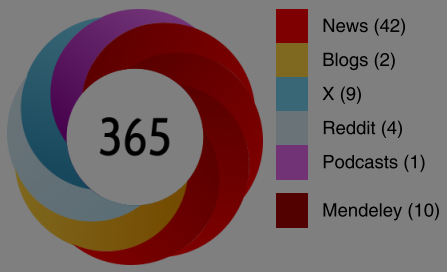
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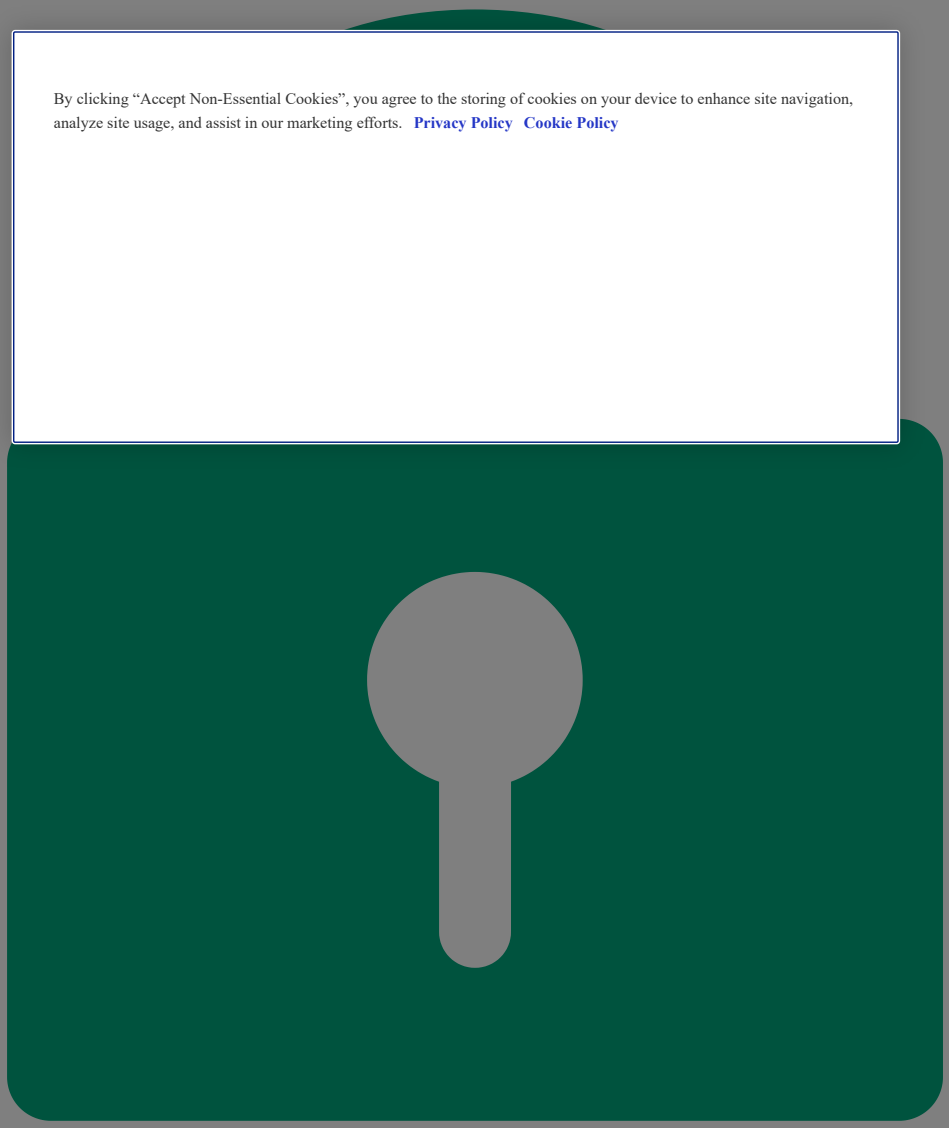
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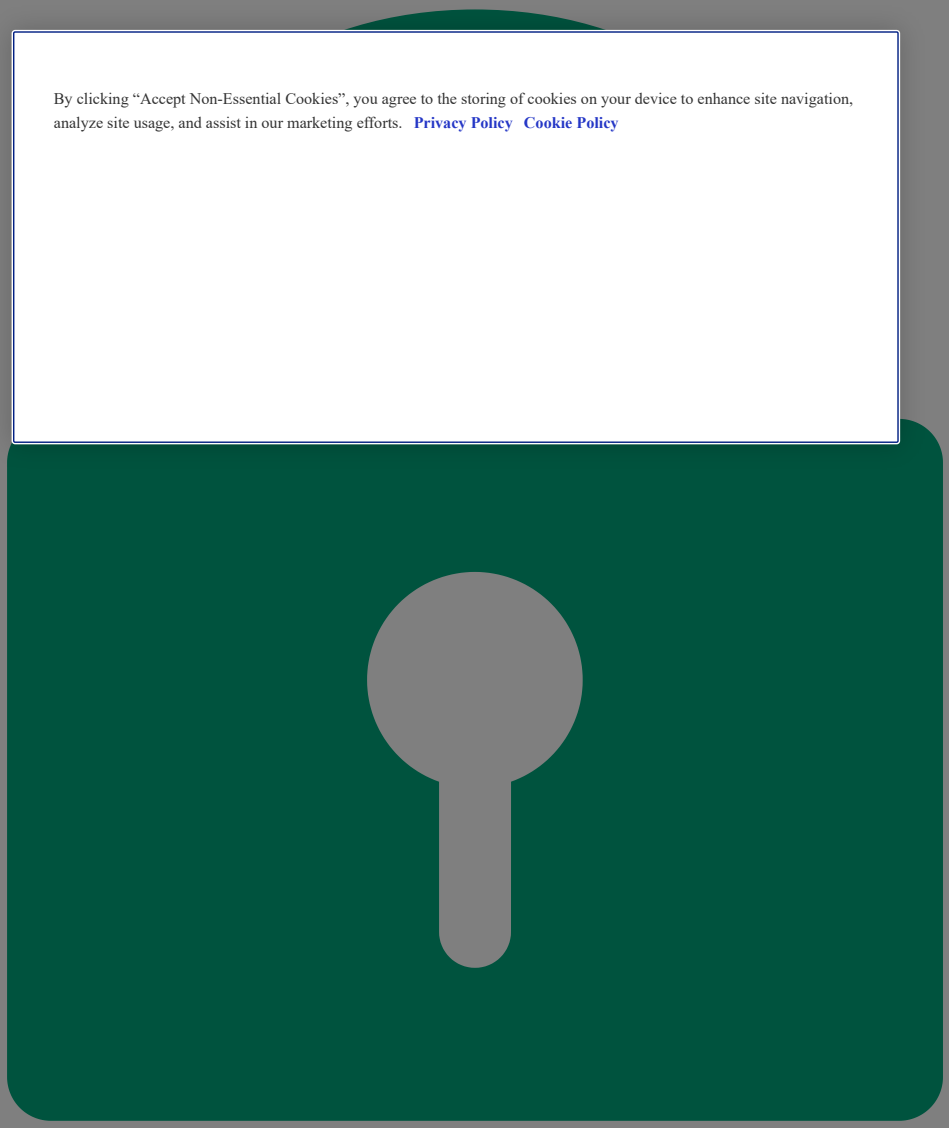
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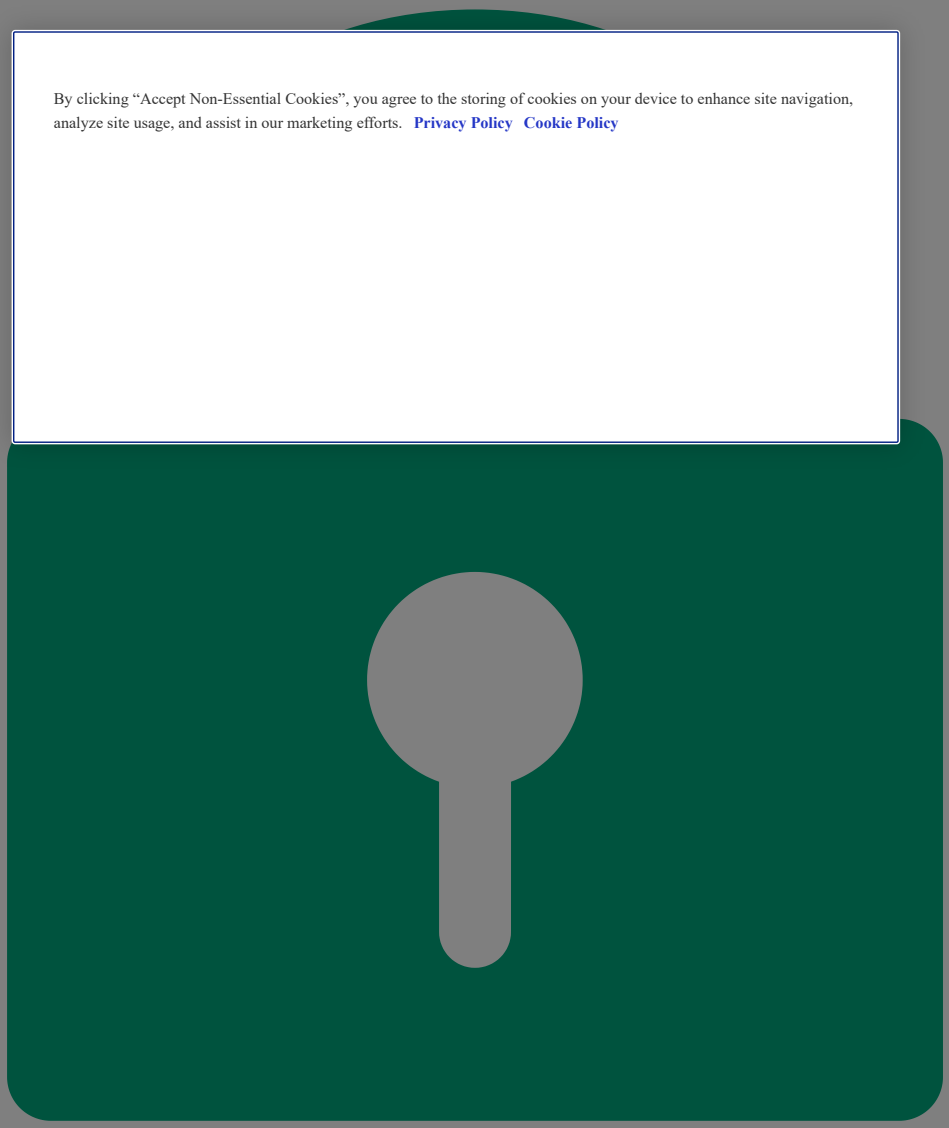
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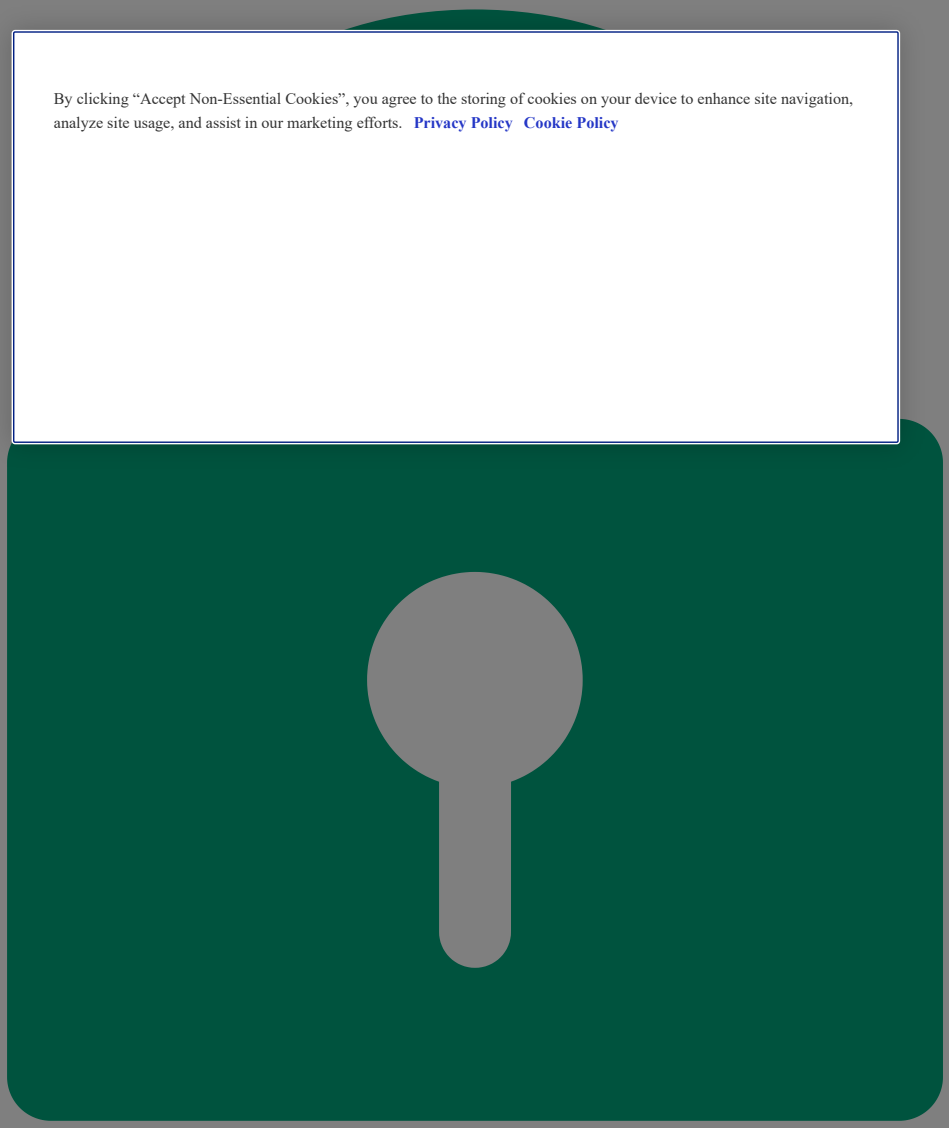
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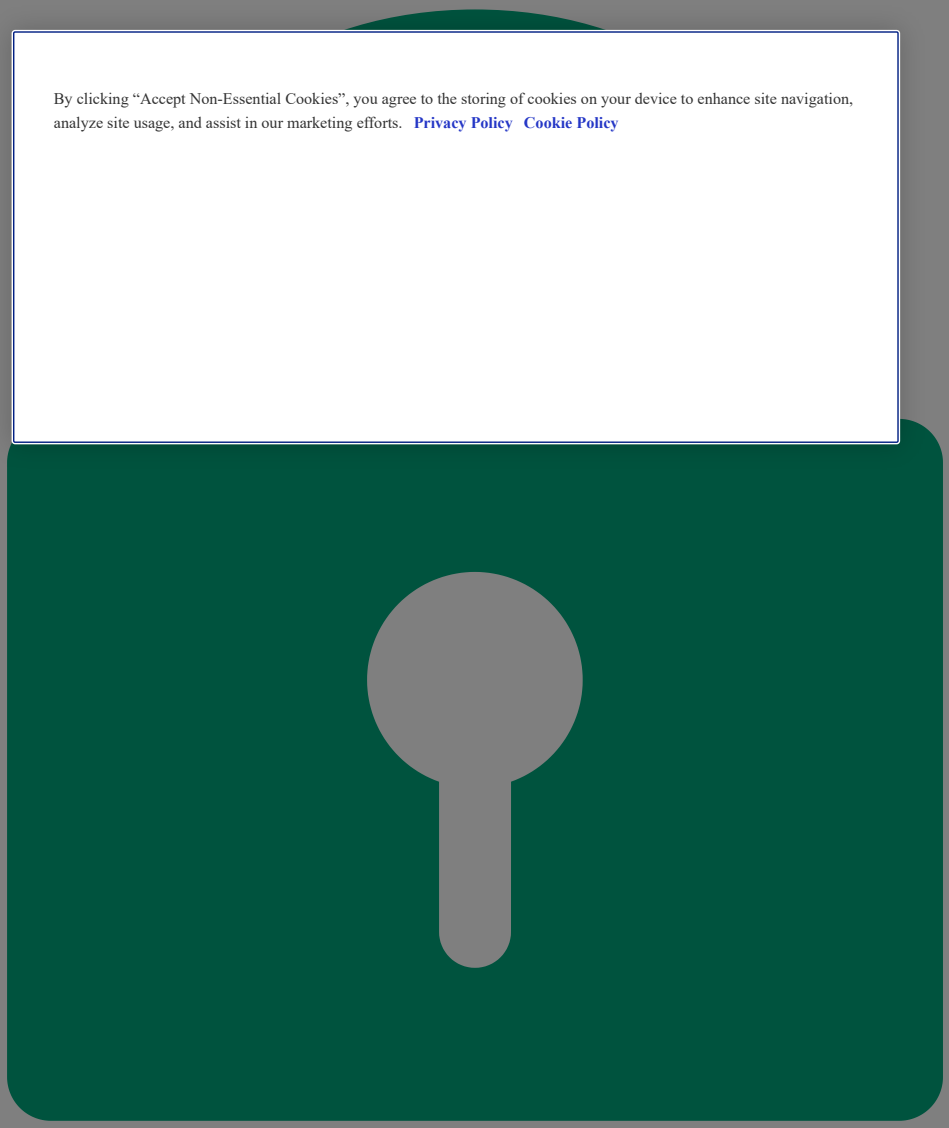
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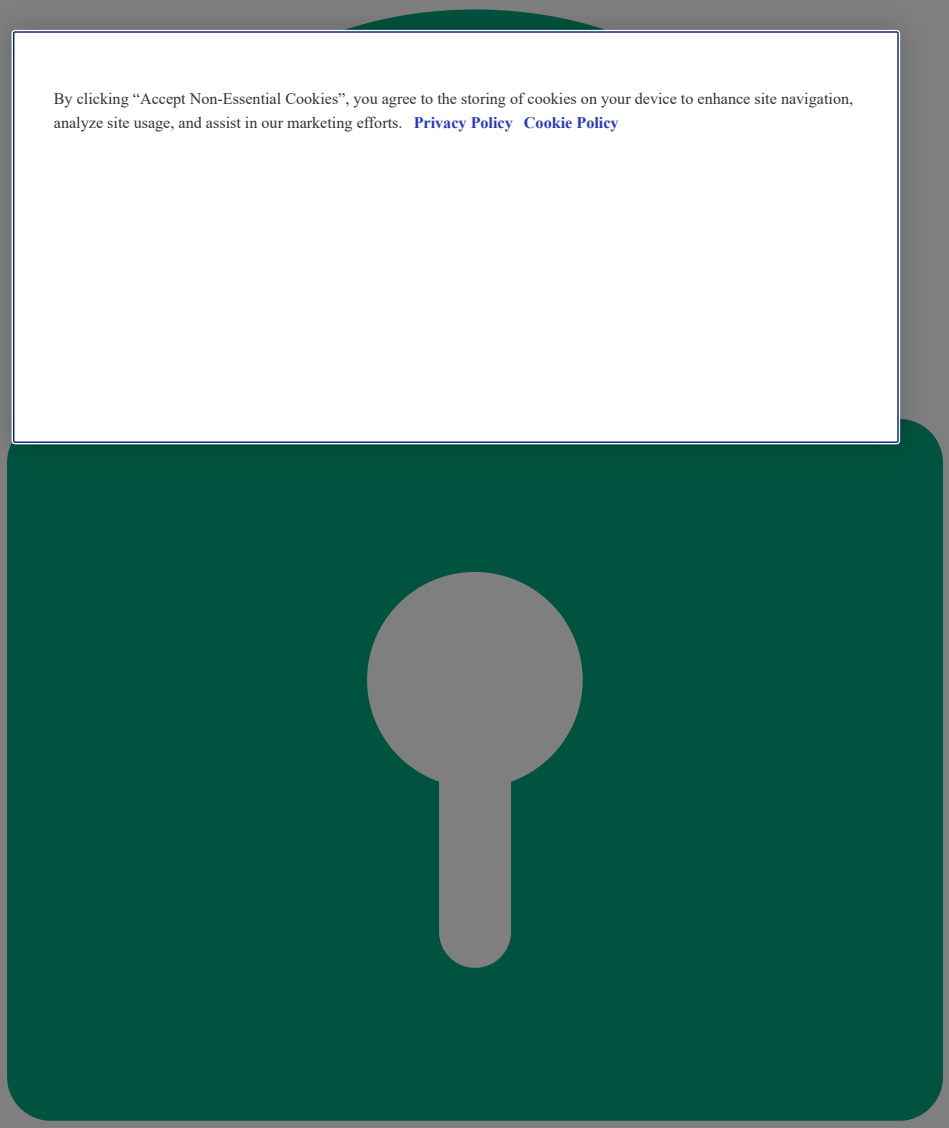
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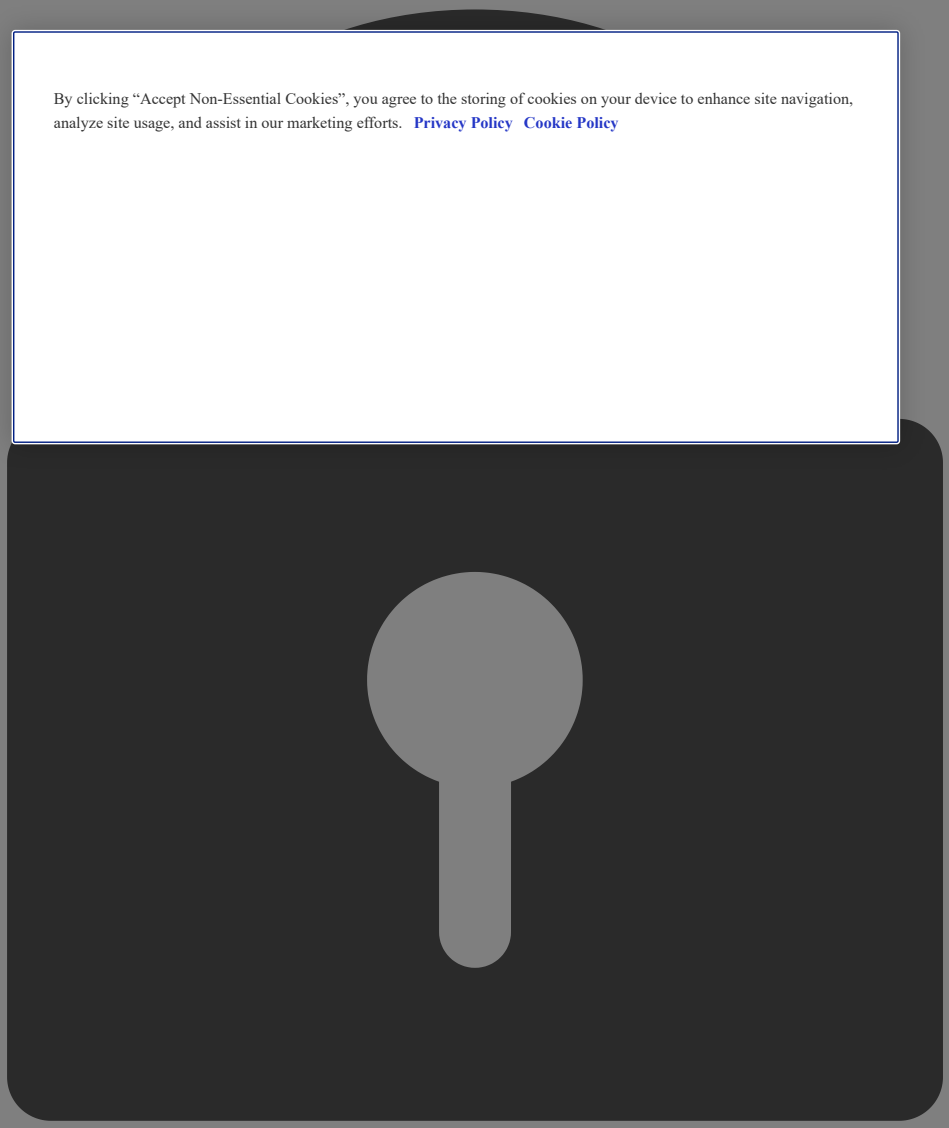
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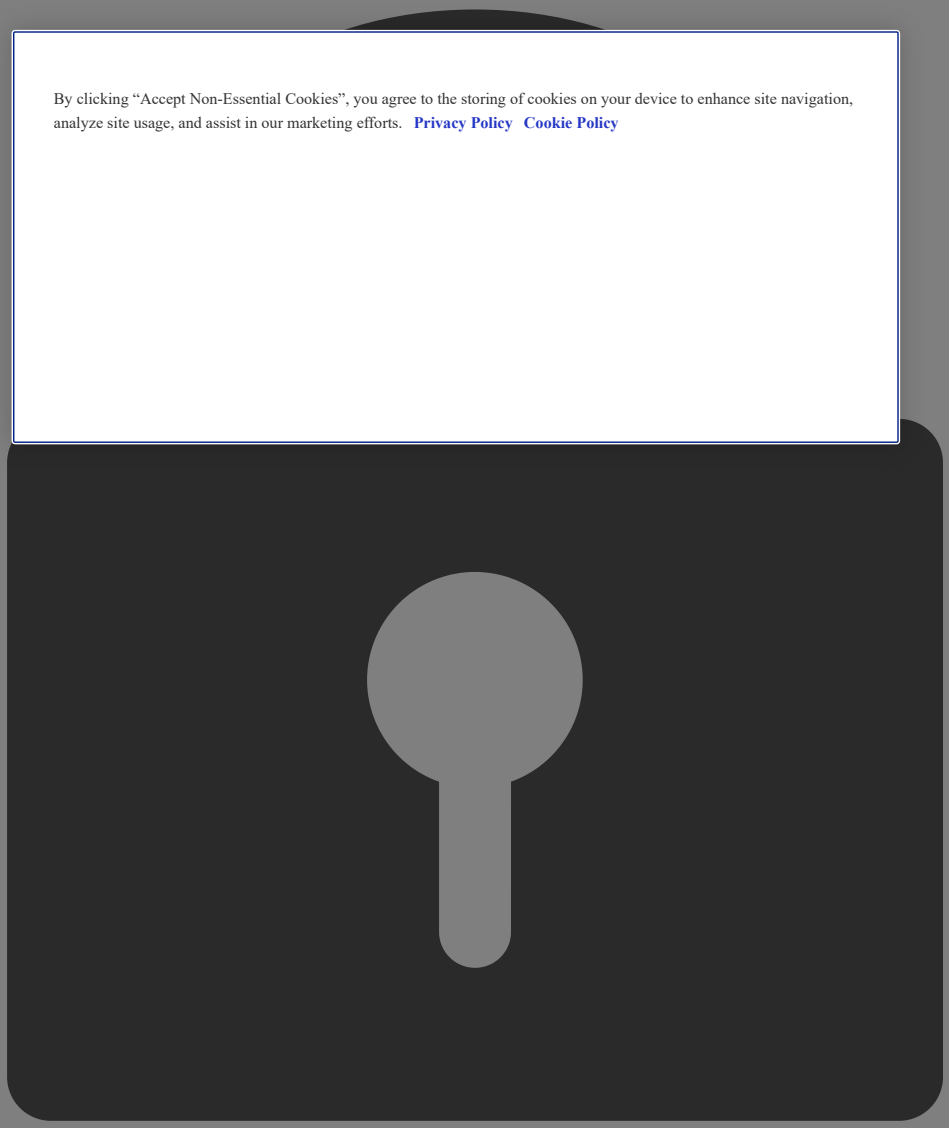
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